



Introduction to Cybersecurity

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Lecture 4

Lecture Rules



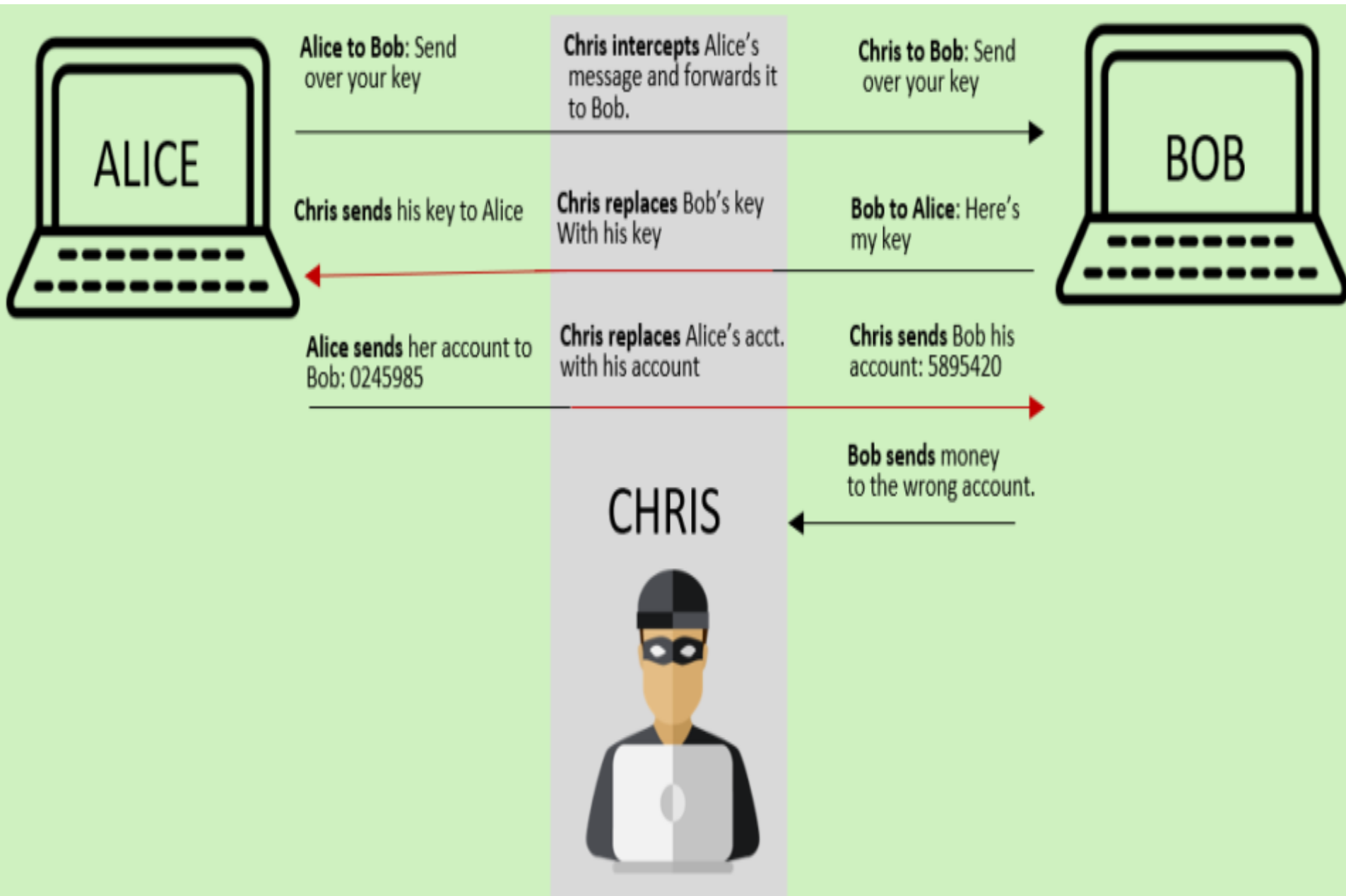
Cryptography Basics

Part I

Lecture Outline

- **Overview**
- **Importance of Secure Communication**
- **Cryptography**
- **Terminology**
- **Cryptographic Services**
- **One-Time Password (OTP) Cipher**
- **Rot13 Cipher**

Overview



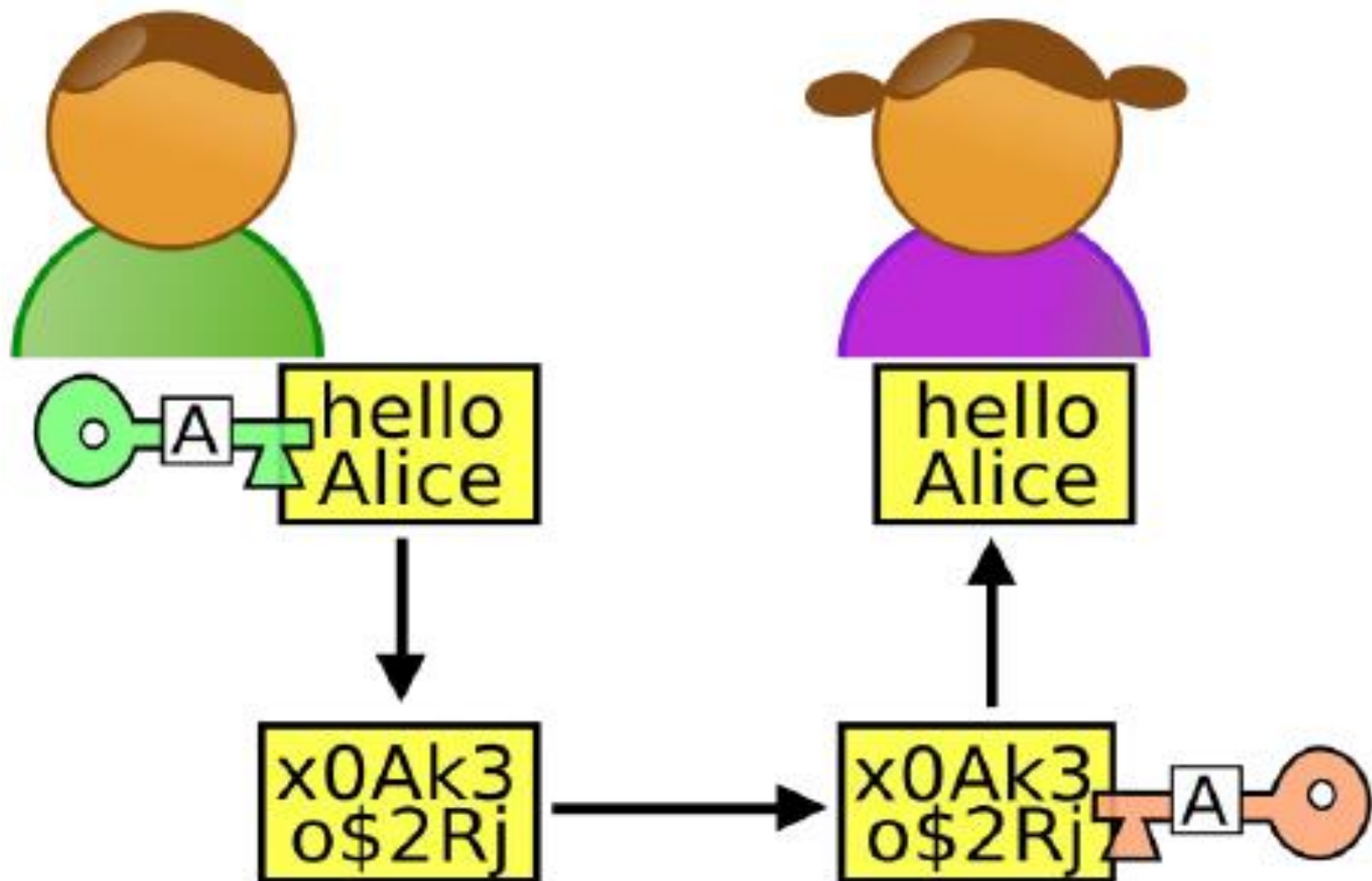
Importance of Secure Communication

- We need **secure communication** in two disciplines:
 - **War time communication.**
 - **Business transactions.**
- Requirements of secure communication (secret message)
 1. **Secrecy**
 - Only the **intended receiver understands** the **message**.
 2. **Authentication**
 - The **sender** and **receiver** need to **confirm** each other's **identity**.
 3. **Integrity**
 - Ensure that their **message** has not been **altered**, **either maliciously** or by **accident** during **transmission**.

Cryptography

- The word cryptography comes from the two Greek words: Krypto (secret) and graphein (write). So cryptography means secret writing .
- The art and science of keeping messages secure is called cryptography and it is practiced by cryptographers
- Modern cryptography exists at the intersection of the disciplines of mathematics, computer science, and electrical engineering. Applications of cryptography include ATM cards, computer passwords, and electronic commerce.

Cryptography as a System



Terminology

- **Plaintext** - original message.
- **Ciphertext** – encoded (**scrambled**) message.
- **Cipher** - algorithm for transforming plaintext to ciphertext.
- **Key** - **some critical information** used by the **cipher**, **known** only to the **sender** and **receiver**.
- **Encipher (encrypt)** - **converting plaintext** to **ciphertext**
- **Decipher (decrypt)** - **recovering plaintext** from **ciphertext**.

Terminology

- **Cryptanalysis (codebreaking)** - study of principles/ methods of decrypting ciphertext **without knowing key**.
- **Cryptology** – field (domain) of both **cryptography** and **cryptanalysis**.

How Encryption Works?



How Decryption Works?

Cipher text

Apply Decryption
Algorithm

Plaintext

D@#%^!245

Decryption

Hi buddy!



Cryptographic Services

- **Confidentiality:** To protect a **user's identity** or data from **being understand or read**.
- **Authentication:** To ensure that message is **verified** from a **authorized** party.
- **Integrity:** To **ensure** that the message **hasn't** been **changed** or **altered** through transmission.
- **Non-repudiation:** To **prevent** an **intended** sender from **denying** that he **sent** a **message**.

Confidentiality

- **Two solutions to achieve confidentiality:**
 - **Encryption.**
 - **Access control.**

Authentication

- Two solutions to ensure authentication are:
 - Passwords
 - Digital signatures

Integrity

- Technical solutions include:
 - Encryption
 - Hashing algorithms

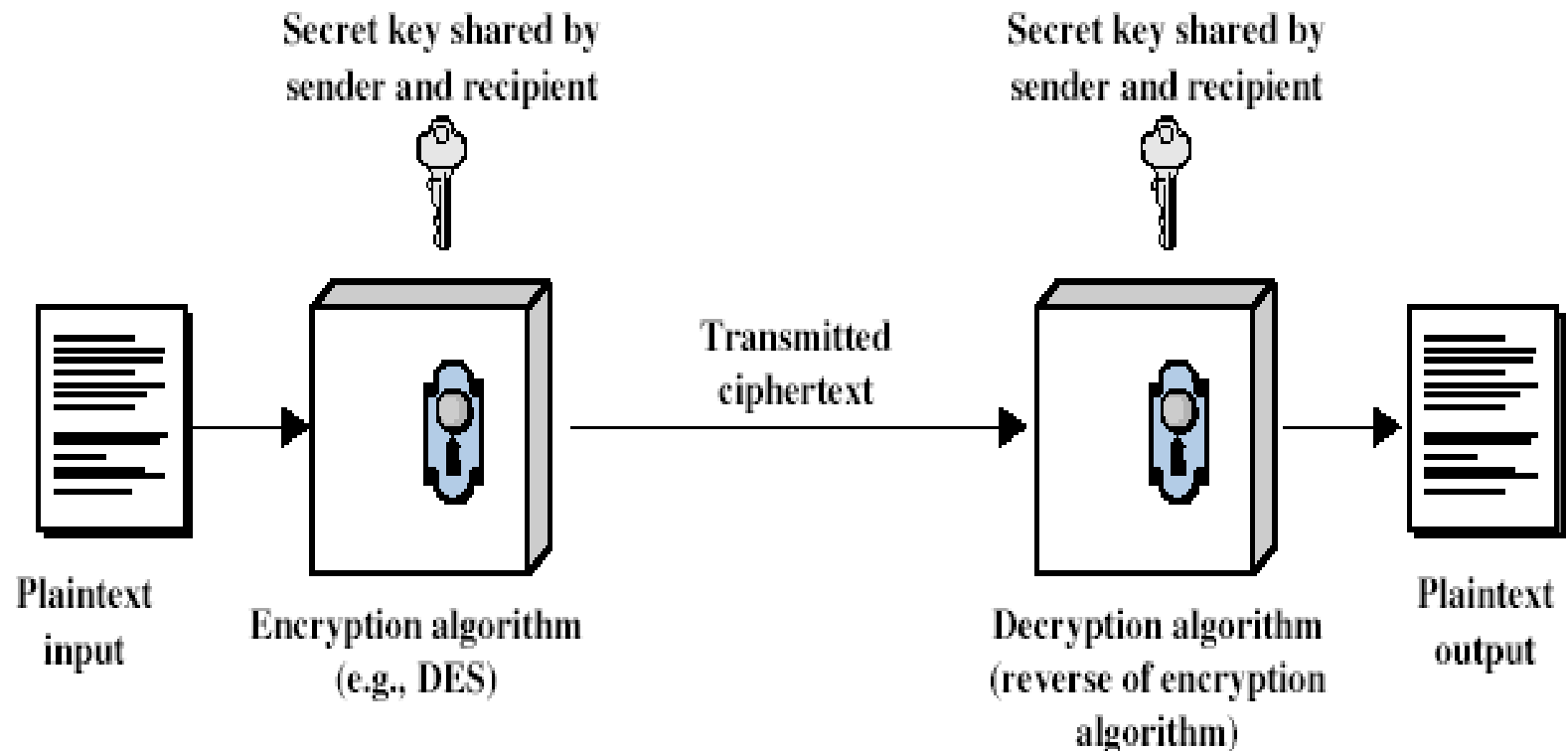
Non-repudiation

- **One** solution to ensure Non-repudiation:
 - Digital signatures

Types of Cryptographic Algorithm

1. **Symmetric Algorithm:** it is also called Secret Key Cryptography (SKC). Uses a single key for both encryption and decryption
2. **Asymmetric algorithm:** it is also called Public Key Cryptography (PKC). Uses one key for encryption and another for decryption
3. **Hash Functions:** Uses a mathematical transformation to irreversibly "encrypt" information

Symmetric Cipher Model



One-Time Password (OTP) Cipher

- **Perfect encryption scheme**, primarily used for **ultra-secure low bandwidth** channels invented in **1917**.
- **The algorithm steps are as follows:**
 1. Start by representing the message as a sequence of **0s** and **1s**
 2. Key is a random sequence of **0s and 1s** of the same **length** as the **message**.
 3. **Add** the key to the message **mod 2**, **bit** by **bit** (Actually **XOR operation**).
 4. **Discard the key, once used and never use again**

(plaintext) 00101001

(key) + 10101100

(ciphertext) 10000101

Why is OTP Unbreakable?

- The key is as long as the given message.
- The key is **random** and specially **auto-generated**.
- Each key should be **used once** and **destroyed** by **both sender** and **receiver**.
- There should be **two** copies of **key: one** with the **sender** and **other** with the **receiver**.
- OTPs have applications in today's world, primarily for **ultra-secure low-bandwidth** channels.

Variations of OTP Characteristics

- Consider **plaintext** as a **sequence** of **letters**.
- **Key** is used **one time only** and **never used** again for any other **message** to be **encrypted**.
- Key is a **random sequence** of shifts between **0** and **25**.
- **Decryption** uses the **same key**, but **subtracts** instead of **adding** the **shifts**.
- This **encryption** is completely **unbreakable** for a **ciphertext attack**.

Example (1)

- Assume $A \rightarrow 1, B \rightarrow 2, \dots, Z \rightarrow 0$, using OTP to get the **ciphertext** from the following:

Plaintext: ONETIMEPAD, **key:** TBFRGFARFM

Solution:

How? $O(15) + T(20) \bmod 26 = 9 = I$
 $N(12) + B(2) \bmod 26 = 14 = P$
 $E(5) + F(6) \bmod 26 = 11 = K$
 $T(20) + R(18) \bmod 26 = 12 = L$
 $I(9) + G(7) \bmod 26 = 16 = P \dots$, etc.

So, Ciphertext: IPKLPSFHGQ

Example (2)

- Assume $A \rightarrow 0, B \rightarrow 1, \dots, Z \rightarrow 25$, using OTP to get the **ciphertext** from the following:
 - Plaintext: **Hello**, Key: **XMCKL**

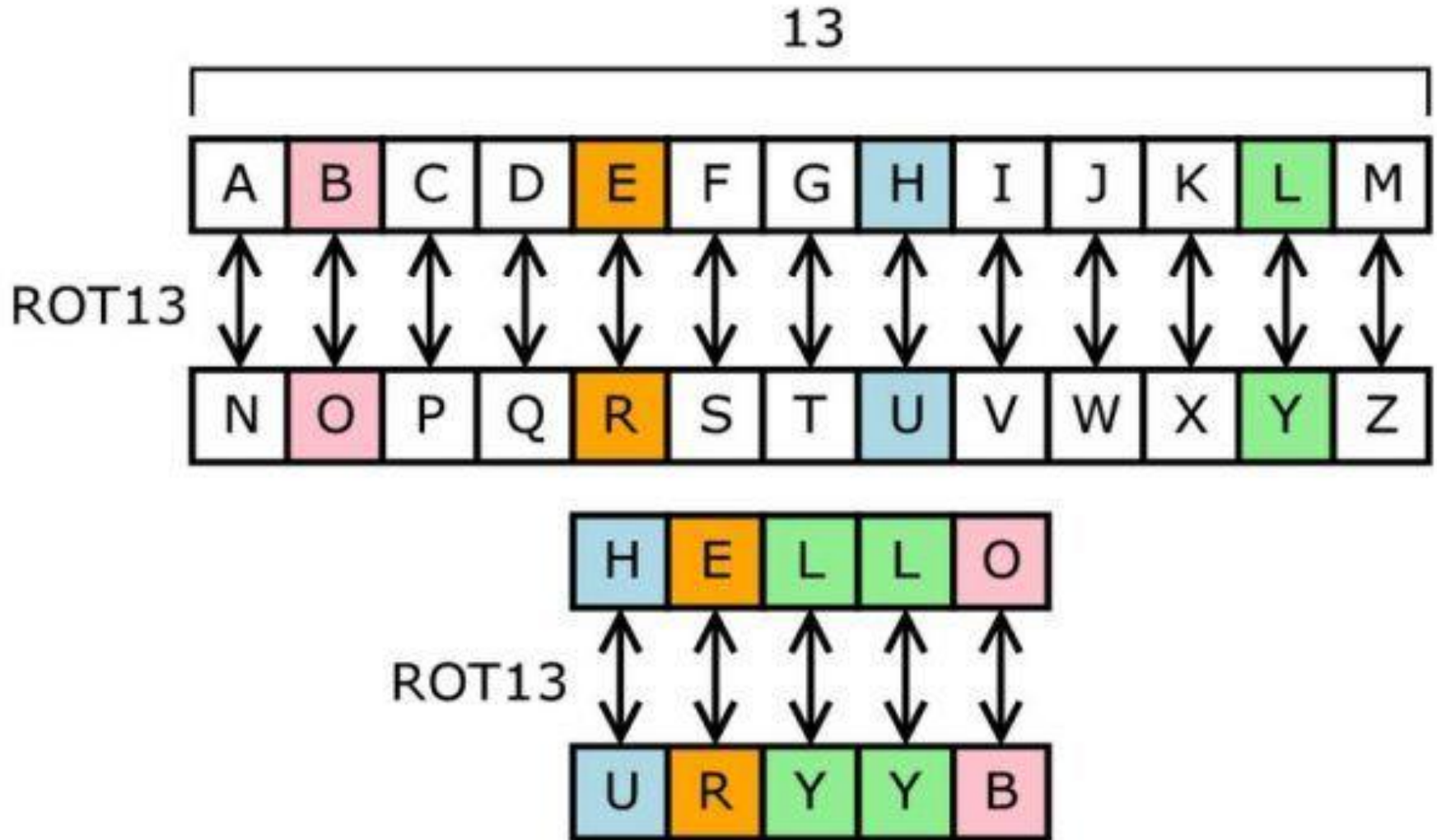
H	E	L	L	O	Message
7	4	11	11	14	
X	M	C	K	L	Key
+ 23	12	2	10	11	
= 30	16	13	21	25	Message + key
= 4	16	13	21	25	Message+key(mod 26)

E Q N V Z $\rightarrow$ ciphertext

Rot13 Cipher

- **ROT13** (short for "**rotate** by **13 places**") is a simple substitution cipher that replaces each letter in a text with the letter **13 positions** down the **alphabet**.
- Because the alphabet has **26 letters**, shifting by **13** and then by another **13** brings you back to the **original letter**.
- **ROT13** only affects letters (**A-Z**).
- While, **Numbers, spaces, punctuation,** and other **characters(\$,&,%,#, ... etc)** are **left unchanged**.
- So that, the same process is used for both **encryption and decryption**.

Rot13 Cipher



Rot13 Cipher

- Example 1 :
- Basic Encryption and Decryption
 - **Plaintext:** "HELLO"
 - **Encryption:**
 - H -> U (Shift by 13)
 - E -> R
 - L -> Y
 - L -> Y
 - O -> B
 - **Ciphertext:** "URYYYB"
 - **Decryption:** Applying **ROT13** again to "**URYYYB**" gives you "HELLO"

Rot13 Cipher

- Example 2 :

- **Plaintext:** "Hello 123!"
- **Ciphertext:** "URYYB 123!"
- **Encryption:**
 - H -> U (**Shift by 13**)
 - E -> R
 - L -> Y
 - L -> Y
 - O -> B
- **Ciphertext:** "URYYB 123!"

Cryptosystem

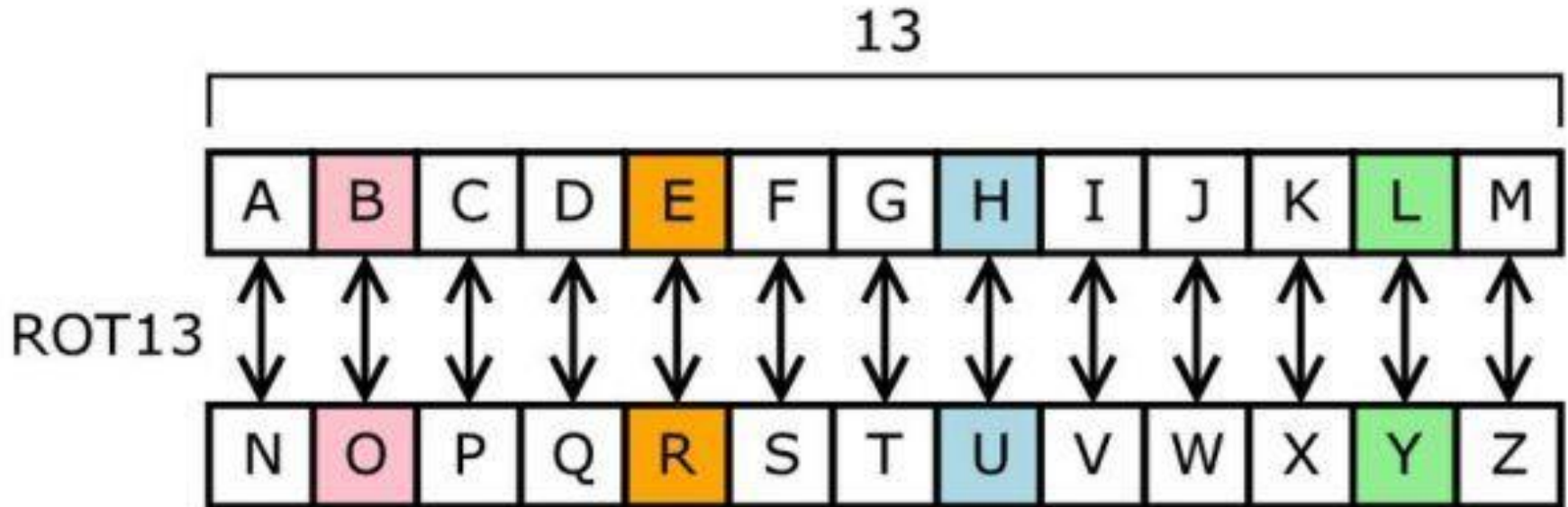
- Cryptosystem – The combination of algorithm, plaintext, ciphertext, key, and key management functions used to perform cryptographic operations is called cryptosystem
- A cryptosystem is a 5-tuple (E, D, M, K, C) , where
 - $E: M \times K \rightarrow C$ —the set of encryption functions;
 - $D: C \times K \rightarrow M$ —the set of decryption functions;
 - M —a set of plaintexts (some use P as symbol);
 - K —the set of keys; **Keyspace**, is a finite set of possible keys.
 - C —the set of ciphertexts;

Task 1 (5 Min)

- Using **Rot13**, decipher the **ciphertext** for the given:
 - **ciphertext**: UVQR GUR ZRFFNTR

Answer

Ciphertext: **UVQR GUR ZRFFNTR**



Plaintext: **Hide the message**

Task 2 (10 Min)

- Assume $A \rightarrow 0$, $B \rightarrow 1$, $C \rightarrow 2$, $D \rightarrow 3$. . . , $Z \rightarrow 25$,

Using **OTP** encryption, find the **ciphertext** for the given **plaintext** and **OTP-key** using the following:

Plain text: T H I S I S S E C R E T

OTP-Key : X V H E U W N O P G D Z

Answer

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Plain text: T H I S I S S E C R E T
OTP-Key : X V H E U W N O P G D Z

Ciphertext: Q C P W C O F S R X H S

thanks for
listening!